Data analysis

Research Methods in AI

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Data analysis?

Not for today: data integration, cleaning, preprocessing

Also not: qualitative data analysis, text mining, deep learning, LLMs

But we will discuss:

- Exploratory analysis
- Confirmatory analysis (estimation and testing)
- Prediction (mostly classification)

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Exploratory analysis

The data from last week

We have a fake chatbot dataset with:

- Emojis: whether emojis were used (yes/no)
- Response length: number of words in responses
- Latency: total time to respond in seconds
- Great question: number of times the chatbot responded with "Great question"
- Frustration: how frustrated the user feels (0-10)

Exploratory data analysis

In exploratory data analysis, we look at patterns in the data, without modelling or testing them:

- Distributions of single variables
- Relationships between variables (descriptive)
- Broader patterns (dimension reduction, clustering)

Descriptives for categorical variables

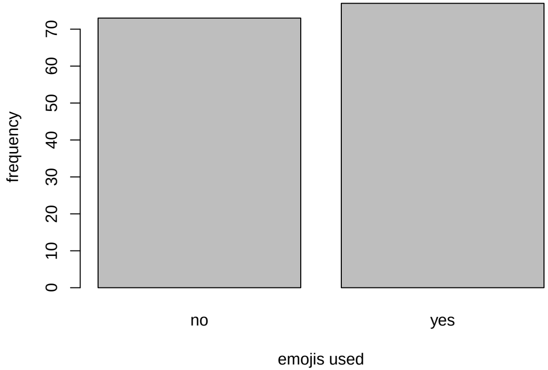
```
> table(emojis)

emojis
 no yes
 73  77

> prop.table(table(emojis))

emojis
       no       yes
0.4866667 0.5133333
```

The mode is the most common value

So yes in this case

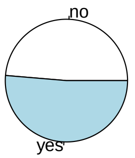
Plots for categorical variables

```
> barplot(table(emojis), xlab = "emojis used", ylab = "frequency")
```



Plots for categorical variables

```
> pie(table(emojis))
```



Pie charts are sometimes mocked by statisticians. For a decent discussion, see:

`https://clauswilke.com/dataviz/visualizing-proportions.html`

Descriptives for numerical variables

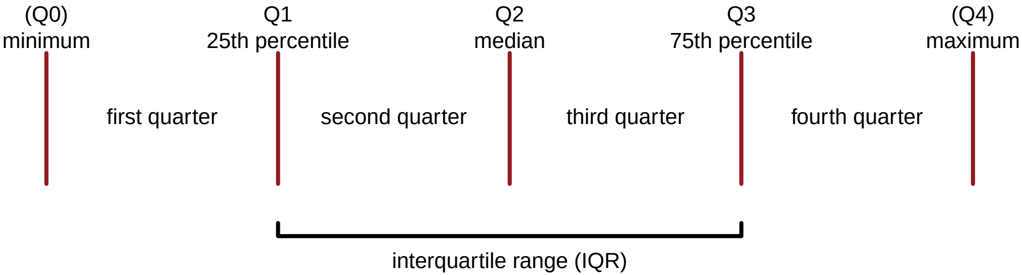
```
> mean(latency)

[1] 2.030961

> median(latency)

[1] 1.973198

> summary(latency)

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.5707  1.6386  1.9732  2.0310  2.4340  3.3982
```

The mean is average value, the median the value of the average observation

And `summary()` gives the five-number summary (quartiles) and the mean

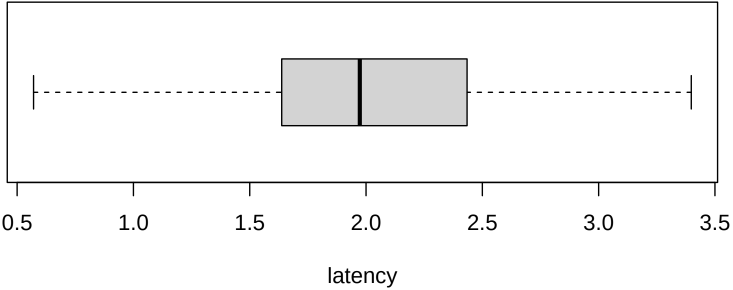
Descriptives for numerical variables

Quartiles divide the data points into four equal parts (the 25% of values each): `summary()`



Plots for numerical variables

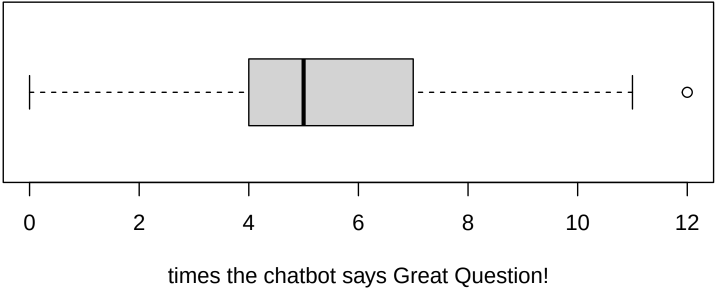
```
> boxplot(latency, xlab = "latency", horizontal = TRUE)
```



A boxplot shows the values of the quartiles

Plots for numerical variables

```
> boxplot(great_question, xlab = "times the chatbot says Great Question!", horizontal = TRUE)
```



… unless we have outliers (> 1.5 times IQR outside of the box)

Descriptives for numerical variables

The variance (s^2) is the square of the standard deviation (s):

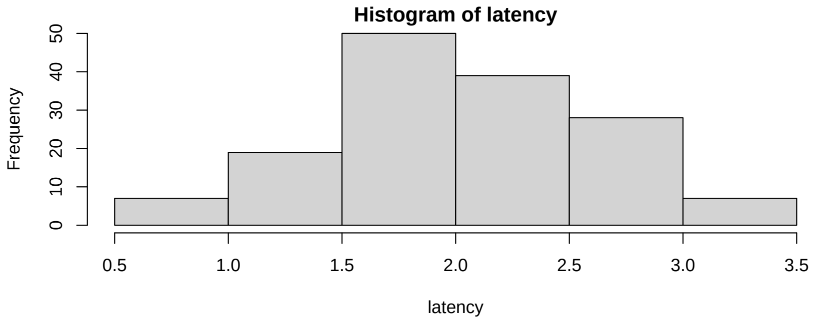
the variance and standard deviation in a sample

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}(x_i - \overline{x})^2\right) \qquad s = \sqrt{s^2}$$

```
> var(latency)

[1] 0.3193959

> sd(latency)

[1] 0.5651513
```

Plots for numerical variables

```
> hist(latency)
```



A histogram gives a better idea of the shape of a distribution

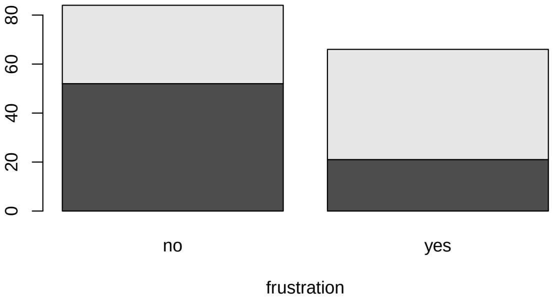
Categorical vs. categorical

As descriptive statistics, for two categorical variables we usually make a cross-table:

```
> table(emojis, frust_binary)

      frust_binary
emojis no yes
   no  52  21
   yes 32  45
```

Categorical vs. categorical

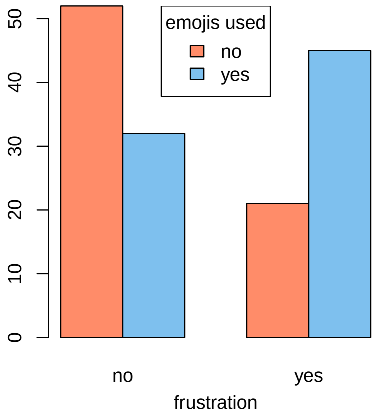
```
> barplot(table(emojis, frust_binary), xlab = "frustration")
```



We can make a nicer version of this barplot...

Categorical vs. categorical

```
> barplot(table(emojis, frust_binary),
+         beside = TRUE, xlab = "frustration",
+         col = c("salmon1", "skyblue2"))
>
> legend("top", title = "emojis used",
+        legend = c("no", "yes"),
+        fill = c("salmon1", "skyblue2"))
```



Numerical vs. numerical

The relationship between numerical variables is often described with a correlation:

```
> cor(latency, response_length)  # Pearson (default)
[1] -0.06438707
> cor(latency, response_length, method = "spearman")
[1] -0.08432256
```

The Spearman correlation uses the ranks instead of the actual values, which makes it more robust

Numerical vs. numerical

the variance and standard deviation in a sample

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}(x_i - \overline{x})^2\right) \qquad s = \sqrt{s^2}$$

the covariance between x and y

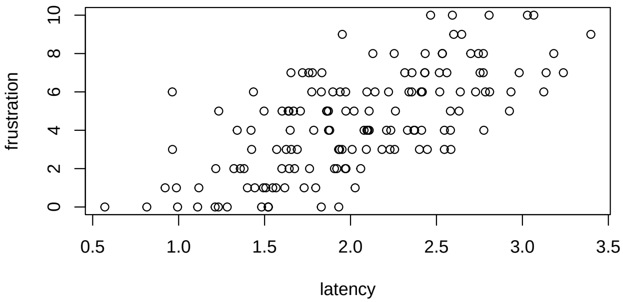
$$\mathrm{cov}_{x,y} = s_{xy} = \frac{1}{n-1}\left(\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})\right)$$

the Pearson correlation between x and y

$$\mathrm{r}_{x,y} = \frac{\mathrm{cov}_{x,y}}{s_x \cdot s_y}$$

Numerical vs. numerical

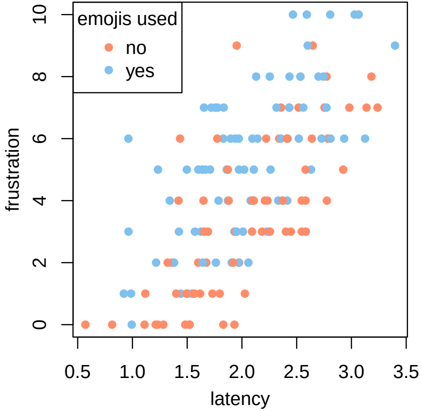
```
> plot(latency, frustration)
```



Let's add a third variable (`emojis`) and make it fancy...

Numerical vs. numerical

```
> colvec <- c("salmon1", "skyblue2")
>
> emoji_cols <- colvec[(emojis == "yes") + 1]
>
> plot(latency, frustration,
+      pch = 16,
+      col = emoji_cols)
>
>
> legend("topleft", title = "emojis used",
+        legend = c("no", "yes"),
+        pch = 16,
+        col = colvec)
```



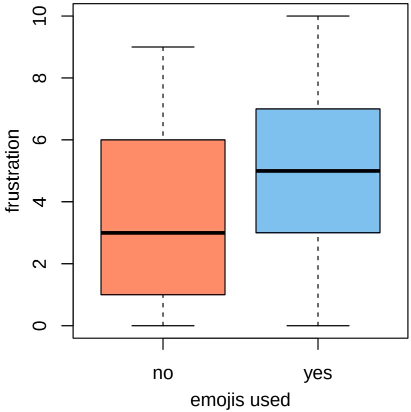
Categorical vs. numerical

In R we can calculate descriptive statistics split up by group:

```
> by(frustration, emojis, summary)

emojis: no
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   1.000   3.000   3.534   6.000   9.000
------------------------------------------------------------
emojis: yes
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   3.000   5.000   5.221   7.000  10.000

> # tapply(frustration, emojis, summary) # alternative
```

Categorical vs. numerical

```
> colvec <- c("salmon1", "skyblue2")
>
> boxplot(frustration ~ emojis,
+       xlab = "emojis used", ylab = "frustration",
+       col = colvec)
```



More than two variables

Plots with more than 2 dimensions don't work very well

We can use dimension reduction methods:

- Linear example: principal component analysis (PCA, in R `prcomp()`)
- Non-linear examples: t-SNE (`Rtsne::Rtsne()`), UMAP (`umap::umap()`)

More than two variables

An alternative way to find structure in a dataset is clustering (a.k.a. unsupervised classification)

Examples:

- Agglomerative hierarchical clustering (`hclust()`)
- k-means clustering (`kmeans()`)

Exploration

Looking at your data is essential

You can spot outliers, data entry errors, deviations from model assumptions

It can also give ideas for hypotheses to test

But if you test based on exploration... be careful!

Rest of the lecture

Are the patterns we see just random noise, or is there a meaningful relationship?

Can we test hypotheses?

Can we make predictions?

Confirmatory analysis

Question

Do any of our independent variables:

- Emojis
- Response length
- Latency
- Great question

have an effect on our dependent variable: Frustration?

Fitting a linear model

Ordinary least squares (OLS) regression in R:

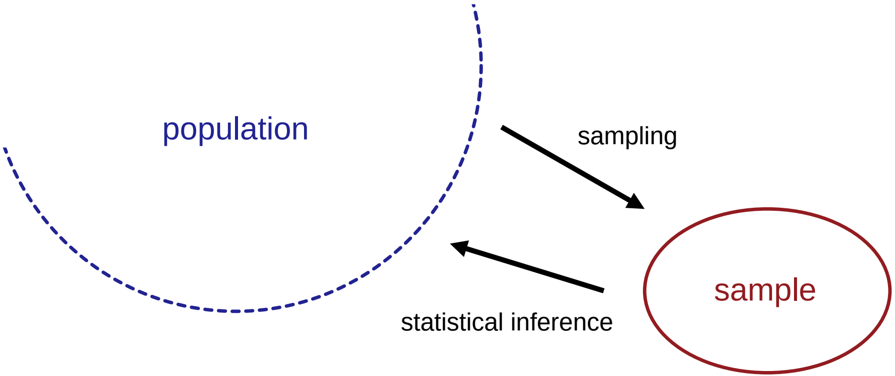
```
> lm(frustration ~ latency + great_question + response_length + emojis)


Call:
lm(formula = frustration ~ latency + great_question + response_length +
    emojis)

Coefficients:
    (Intercept)          latency   great_question  response_length
      -2.574657         3.049756         0.158479        -0.008188
       emojisyes
        1.718356
```

Remember our model for simulation: $y = -2 + 3 \cdot x_{la} + .1 \cdot x_{gc} - .01 \cdot x_{rl} + 2 \cdot \mathrm{I}(x_{em} = \text{"yes"}) + \epsilon$

Fitting a linear model

```
> out_lm <- lm(frustration ~ latency + great_question + response_length + emojis)
> summary(out_lm)


Call:
lm(formula = frustration ~ latency + great_question + response_length +
    emojis)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9724 -1.2173 -0.1764  1.0166  4.7706

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -2.574657   0.707829  -3.637 0.000382 ***
latency          3.049756   0.249589  12.219  < 2e-16 ***
great_question   0.158479   0.061685   2.569 0.011204 *
response_length -0.008188   0.003004  -2.726 0.007209 **
emojisyes        1.718356   0.281674   6.101 9.15e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.712 on 145 degrees of freedom
Multiple R-squared:  0.5929, Adjusted R-squared:  0.5817
F-statistic:  52.8 on 4 and 145 DF,  p-value: < 2.2e-16
```

Inferential statistics



What are the hypotheses?

```
> out_lm <- lm(frustration ∼ latency + great_question + response_length + emojis)
> summary(out_lm)


Call:
lm(formula = frustration ~ latency + great_question + response_length +
    emojis)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9724 -1.2173 -0.1764  1.0166  4.7706

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -2.574657   0.707829  -3.637 0.000382 ***
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great_question   0.158479   0.061685   2.569 0.011204 *
response_length -0.008188   0.003004  -2.726 0.007209 **
emojisyes        1.718356   0.281674   6.101 9.15e-09 ***
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Residual standard error: 1.712 on 145 degrees of freedom
Multiple R-squared:  0.5929, Adjusted R-squared:  0.5817
F-statistic:  52.8 on 4 and 145 DF,  p-value: < 2.2e-16
```

What are p-values?

p-value

The probability of finding this test statistic, or a more extreme one, if the null-hypothesis is true.

Why don't we just calculate the probability the hypothesis is true/false?

Well... what is a probability?

Quiz

What is the correct definition of probability?

- A Something that is shared equally between all the possible outcomes
- B A relative frequency over time
- C A measure of the 'degree of belief' of the individual assessing uncertainty
- D A physical propensity/disposition/tendency of a given type of physical situation to yield an outcome of a certain kind

Subjective probability in statistics

- The basis of Bayesian statistics
- Probabilities for anything (models, samples, parameters, ...)
- Uses prior and posterior probabilities: updating beliefs
- I *believe* subjective probability makes a lot of sense, but application is a bit less common (no p-values, but posteriors and Bayes Factors)

Frequentist probability in statistics

- The basis of frequentist statistics
- The truth is fixed, but unknown
- So hypotheses don't have probabilities
- Sampling is random, so sample statistics are random variables
- Most common type of statistics in practice, with p-values

Inferential statistics

Idea of frequentist testing:

- Assume nothing going on (H_0)
- Look at data, calculate a statistic (like mean) and a test statistic (like t)
- What is p-value: the chance of observing this data / statistic (or more extreme)?
- If low: then probably H_0 wasn't true (so something is going on)
- If not low: H_0 could be true
- What is "low"? Threshold often $\alpha = .05$

Errors

But... we use a sample to make a statement about a population, so we could be making a mistake.

Type I error (probability α, yes the same as the threshold!)

Rejecting the null-hypothesis while it is true.

We conclude there's an effect, but that's not true.

Type II error (probability β)

Failing to rejecting a null-hypothesis that is not true.

We miss an effect that is actually present.

Point estimates of effects

If we have uncertainty, perhaps we should not report point estimates

```
> out_lm <- lm(frustration ~ latency + great_question + response_length + emojis)
> out_lm


Call:
lm(formula = frustration ~ latency + great_question + response_length +
    emojis)

Coefficients:
    (Intercept)          latency   great_question  response_length
      -2.574657         3.049756         0.158479        -0.008188
      emojisyes
       1.718356
```

Confidence intervals of effects

… but confidence intervals

```
> out_lm <- lm(frustration ~ latency + great_question + response_length + emojis)
> confint(out_lm)

                      2.5 %       97.5 %
(Intercept)     -3.97365264 -1.175660874
latency          2.55645446  3.543058092
great_question   0.03656080  0.280398127
response_length -0.01412541 -0.002250514
emojisyes        1.16163923  2.275073615
```

Which number is not included in any of these intervals?

What does that mean?

Be carful

What is the danger of doing exploratory and confirmatory analysis on the same data?

`https://tylervigen.com/spurious-correlations`

Quiz Q1/3

In statistics, what is a type I (type 1) error?

A H_0 is rejected, while H_0 is true

B H_a is rejected, while H_a is true

C H_0 is retained, while H_0 is true

D H_a is retained, while H_a is true

Quiz Q2/3

Practically (perhaps unjustly), the most important numbers in statistics are p-values. Which of the following statements about the p-value is *most* correct?

The p-value is a probability statement about...

A the null-hypothesis

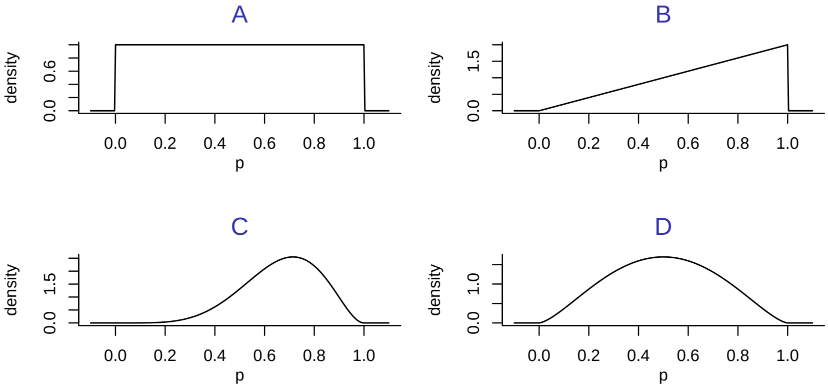
B the alternative hypothesis

C the data

D making type I errors

Quiz Q3/3

If the null-hypothesis is true (so "no pattern in the data"), what should the distribution of the p-value look like?



Other models: logistic regression

```
> out_glm <- glm(frust_binary == "yes" ~ latency + great_question + response_length +
+      emojis, family = "binomial")
> summary(out_glm)


Call:
glm(formula = frust_binary == "yes" ~ latency + great_question +
    response_length + emojis, family = "binomial")

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)    -11.030977   2.475093  -4.457 8.32e-06 ***
latency          5.263605   1.000634   5.260 1.44e-07 ***
great_question   0.488407   0.155760   3.136  0.00171 **
response_length -0.044628   0.008967  -4.977 6.46e-07 ***
emojisyes        4.152705   0.909933   4.564 5.02e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 205.779  on 149  degrees of freedom
Residual deviance:  78.214  on 145  degrees of freedom
AIC: 88.214

Number of Fisher Scoring iterations: 7
```

Other models: chi-squared test

```
> table(frust_binary, emojis)

            emojis
frust_binary no yes
         no  52  32
         yes 21  45

> chisq.test(frust_binary, emojis)


 Pearson's Chi-squared test with Yates' continuity correction

data:  frust_binary and emojis
X-squared = 12.215, df = 1, p-value = 0.0004741
```

Other models: independent t-test

```
> t.test(frustration ~ emojis)


 Welch Two Sample t-test

data:  frustration by emojis
t = -4.0947, df = 145.83, p-value = 6.977e-05
alternative hypothesis: true difference in means between group no and group yes is not equal to 0
95 percent confidence interval:
 -2.5005591 -0.8725062
sample estimates:
 mean in group no mean in group yes
         3.534247          5.220779
```

Prediction

Question

Can we predict our label: Frustration (yes/no)

From our features:

- Emojis
- Response length
- Latency
- Great question

Models for testing and prediction

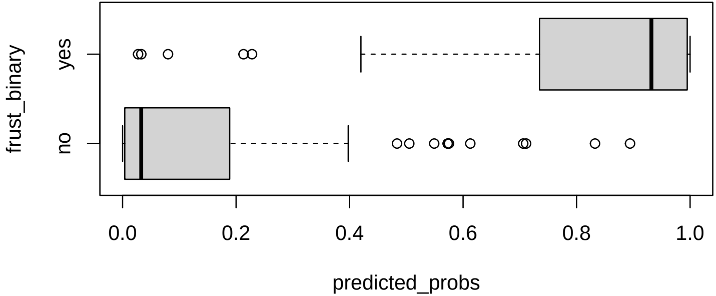
Note that models for inference, like (logistic) regression, can also be used for prediction!

Other models are just used for prediction. For categorical outcomes, some common ones are random forest, k-nearest neighbours, and neural net

These examples are more black box: it is hard to say *why* a prediction is made

Prediction with logistic regression

```
> out_glm <- glm(frust_binary == "yes" ~ latency + great_question + response_length +
+     emojis, family = "binomial")
> predicted_probs <- predict(out_glm, type = "response")
> boxplot(predicted_probs ~ frust_binary, horizontal = TRUE)
```



Confusion matrix and accuracy

```
> table(predicted_probs > 0.5, frust_binary)

       frust_binary
        no yes
  FALSE 75   7
  TRUE   9  59
```

```
> confus_mat <- table(predicted_probs > 0.5, frust_binary)
> sum(diag(confus_mat))/sum(confus_mat)

[1] 0.8933333
```

Random forest in R

```
> library(randomForest)
> randomForest(factor(frust_binary) ~ latency + response_length + great_question +
+     emojis)


Call:
 randomForest(formula = factor(frust_binary) ~ latency + response_length +      great_question + emoj
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 17.33%
Confusion matrix:
    no yes class.error
no  72  12   0.1428571
yes 14  52   0.2121212
```

Neural net in R

```
> library(nnet)
> nnet(factor(frust_binary) ~ scale(latency) + scale(response_length) + scale(great_question),
+     size = 5, decay = 0.01)

# weights:  26
initial  value 105.626083
iter  10 value 57.546925
iter  20 value 54.097839
iter  30 value 51.183138
iter  40 value 49.353948
iter  50 value 48.123751
iter  60 value 47.933085
iter  70 value 47.907217
iter  80 value 47.825955
iter  90 value 47.377266
iter 100 value 47.362565
final  value 47.362565
stopped after 100 iterations
a 3-5-1 network with 26 weights
inputs: scale(latency) scale(response_length) scale(great_question)
output(s): factor(frust_binary)
```

k-NN in R

```
> library(caret)
> dataf <- data.frame(frust_binary, latency, response_length, great_question, emojis)
> out_knn <- knn3(factor(frust_binary) ~ latency + response_length + great_question +
+     emojis, data = dataf)
> predicted_class <- predict(out_knn, type = "class")
> table(predicted_class, frust_binary)

               frust_binary
predicted_class no yes
            no  69  18
            yes 15  48
```

k-NN in R

```
> library(caret)
> dataf <- data.frame(frust_binary, latency, response_length, great_question, emojis)
> out_knn <- knn3(factor(frust_binary) ~ latency + response_length + great_question +
+     emojis, data = dataf, k = 10)
> predicted_class <- predict(out_knn, type = "class")
> table(predicted_class, frust_binary)

               frust_binary
predicted_class no yes
            no  63  20
            yes 21  46
```

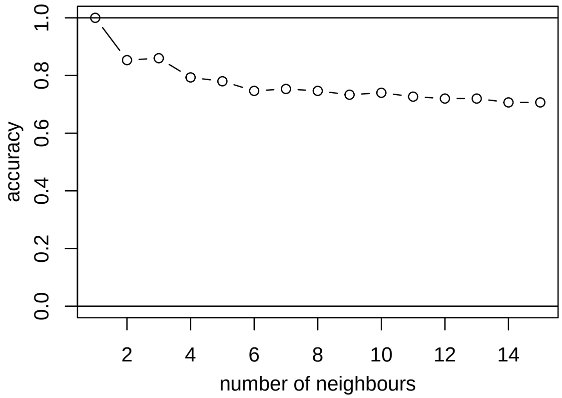
k-NN in R

```
> library(caret)
> dataf <- data.frame(frust_binary, latency, response_length, great_question, emojis)
> out_knn <- knn3(factor(frust_binary) ~ latency + response_length + great_question +
+     emojis, data = dataf, k = 1)
> predicted_class <- predict(out_knn, type = "class")
> table(predicted_class, frust_binary)

               frust_binary
predicted_class no yes
            no  84   0
            yes  0  66
```

Overfitting?

Overfitting



Note that in this case the models on the left (with a smaller number of neighbours) are *more complex*, in the sense that they have a more complex classification boundary.

Increased model complexity tends to give a better fit on the training data, but not necessarily on the test data.

Machine learning

Topics not discussed in this lecture, but assumed to be familiar (Machine Learning course):

- Accuracy
- Sensitivity and specificity
- Class imbalance and F1 scores
- Train, validation, and test sets
- Cross-validation

Conclusion

Summary

We have looked at:

- Exploratory analysis: descriptives and plots
- Confirmatory analysis: hypothesis testing
- Prediction: binary classification

Next week

More classification issues

Replication crisis

Publication bias